

The cloud landscape

Cloud technology is the main force behind digital transformation, but the term itself can often feel abstract. It's common to hear the name without understanding what it truly means. This makes the concept seem like a mystery rather than a real infrastructure.

Cloud Corner: What is the cloud, exactly?

So, before we continue, let's ground the concept of the cloud in reality. Listen in as Katelyn and Joe strip away the jargon to define exactly what the cloud is and how it functions like a modern utility.

Click play to begin. For a transcript, click the option below.

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Transcript

Katelyn: Let's tackle the big one. We hear these terms all the time, but what is the cloud and cloud technology, exactly? And how does it connect to digital transformation?

Joe: That's the foundational question, isn't it? At its core, "the cloud" is really just a simpler, friendlier word for a very complex system. Think about it this way: in order to run an application, you'd need a web of software, servers, computers, and security systems. Instead of listing all that, we've bundled it into one simple word: the cloud.

Katelyn: Exactly. It's not some puffy thing in the sky; it's a metaphor for a huge, global network of data centers. These buildings are filled with powerful computers that store and process information, and you can access all of it right through the internet.

Joe: I kind of think of it like I think of the electrical grid. A century ago, if you wanted to power a factory, you had to build your own generator. Today, you just plug into the wall. The cloud is a lot like that, but for computing power.

Katelyn: That's a perfect analogy. And to better understand how companies "plug in," it helps to explore the different ways they can set up their information technology, or IT infrastructure.

Joe: Right, it's not a one-size-fits-all model. There are a few main approaches you'll hear about, including on-premises, private cloud, public cloud, hybrid cloud, and multicloud.

Katelyn: Say that list five times fast.

Joe: I'm not even going to try.

Katelyn: Now, that might sound like a lot of jargon, but don't worry. In the upcoming sections, we'll examine exactly what each of those implementations means and why you might choose one over the other.

Joe: And it's a perfect time to mention a key point for our learners: when we refer to "the cloud," it's safe to assume we're talking about the public cloud. Again, you'll learn what that term means in a bit, but for immediate reference, Google Cloud is an example of a public cloud.

Katelyn: Excellent clarification.



IT infrastructure models

Now that you're familiar with a basic definition of the cloud, let's explore some common terms associated with IT infrastructure: on-premises, private cloud, and public cloud.

On-premises

On-premises IT infrastructure, or on-prem, refers to hardware and software applications that are hosted on-site, located and operated within an organization's data center to serve their unique needs. This implementation is the traditional way of managing IT infrastructure.

Benefits



On-premises infrastructure operates without third-party access, granting owners full physical control over their server hardware and software. This model also eliminates the need to pay for ongoing access.

Other considerations



Maintenance requirements: To have the computing power to run required workloads, organizations must buy physical servers and other infrastructure through procurement processes that can take months. These systems require physical space, typically a specialized room with sufficient power and cooling. After configuring and deploying the systems, businesses then need expert personnel to manage them. This long process is difficult to scale when demand spikes or business expands.

Over- or underestimating demand: Organizations often acquire more or less computing resources than they actually need, which results in inefficient utilization and high overhead. Cloud computing addresses these issues by offering computing resources as scalable, on-demand services.



Private cloud

Private cloud is a type of cloud computing where the infrastructure is dedicated to a single organization instead of the general public. This type is also known as single-tenant or corporate cloud. Typically, an organization has to perform the same kind of ongoing maintenance and management for a private cloud as it would for traditional on-premises infrastructure. A private cloud is hosted within an organization's own private servers, either at an organization's own data center, at a third-party colocation facility, or by using a private cloud provider.

Benefits



Private cloud computing gives businesses many of the benefits of a public cloud—including self-service, scalability, and elasticity—with more customization available from dedicated on-premises infrastructure.

Other considerations



Organizations might use private cloud if they have already made significant investments in their own infrastructure or if, for regulatory reasons, data must be kept on-premises or hosted in a certain way.



Public cloud

In a **public cloud** deployment model, on-demand computing services and infrastructure are managed by a third-party provider, such as Google Cloud, and shared with multiple organizations or “tenants” through the public internet. This sharing is why public cloud is

known as multi-tenant cloud infrastructure, but each tenant's data and applications running in the cloud are hidden from other tenants.

Benefits

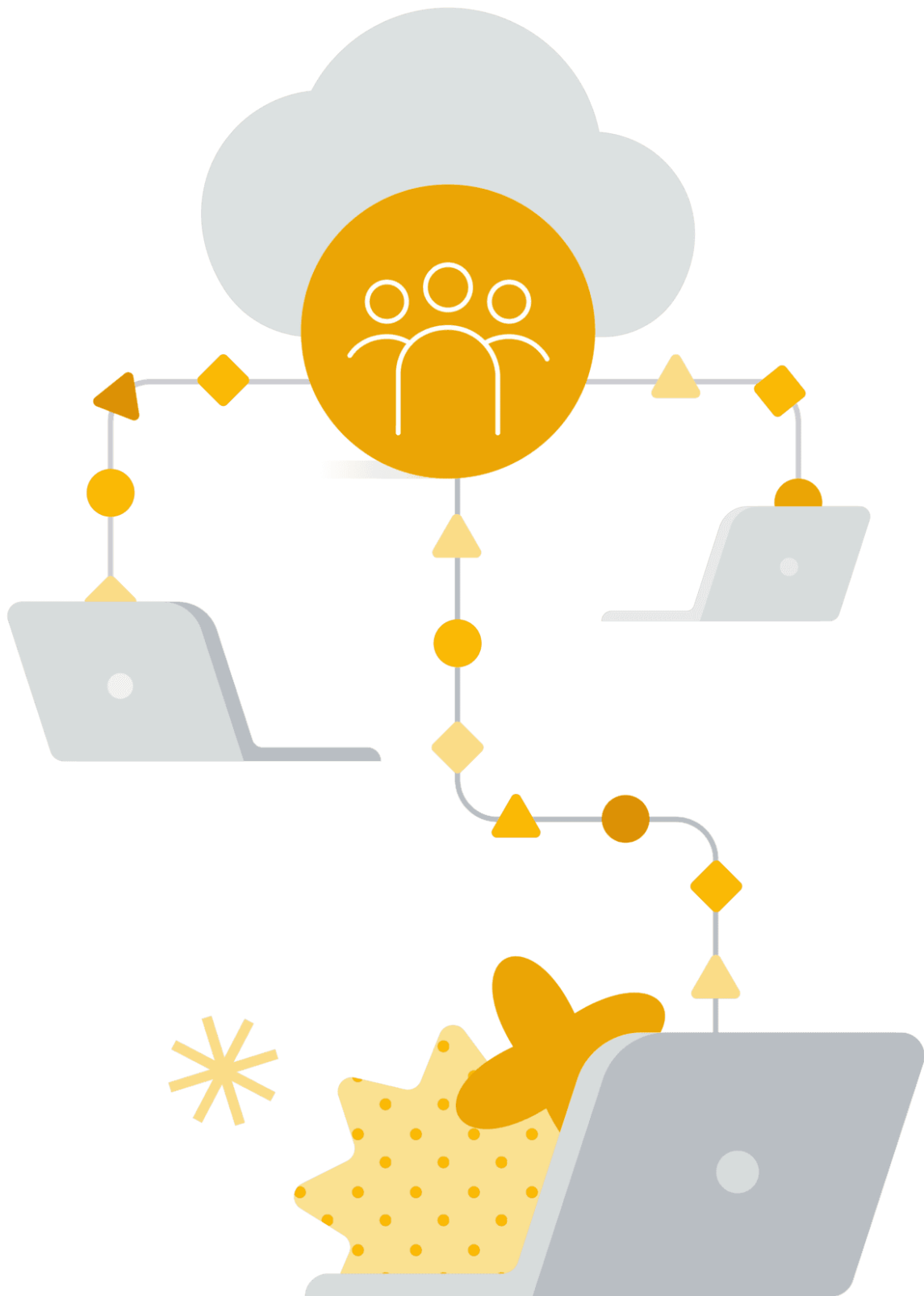


Because the public cloud has on-demand availability of computing and infrastructure resources, organizations don't need to acquire, configure, or manage those resources themselves, and they only pay for what they use.

Other considerations



Think of the public cloud like an apartment building that's maintained by a property management company. The building has many units and tenants. Each unit might have a slightly different layout, but still has all the amenities a tenant needs to live there. And each unit is locked and private to the tenant who pays for that space.



Knowledge check

Drag each card to the correct infrastructure model.

Public cloud

Private cloud

On-premises



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